

Manual

WEP Data Archiving WEP-ARC 8.2

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1 WEP Data Archiving - Overview

Purpose

The function package Archiving Inspection Data from WEP provides access to archived data and additionally to export recorded data and to load these data back into the system as needed.

Use this function package to:

- access data already moved to archive tables using this function package;
- ensure that the data entered is stored long-term to comply with legal regulations or customer requirements;
- import data back into the system for reevaluation or reanalysis to comply with legal regulations or customer requirements;

Integration

Archiving inspection requirements including all referenced inspection data is related to this component

- Inspection planning for goods receipt inspection
- Goods receipt inspection evaluations
- Standard control charts and histograms
- Advanced control charts and histograms and
- Failure mode analysis / action tracking

Features

- The following data areas / objects are archived:
 - Inspection Requirements
 - Inspection Steps
 - Inspection Points
 - Inspection data, e.g. measured values
 - Allocated errors
- Export function
 - Functions to transfer (export) data from the archive tables to external file systems for long-term storage of the recorded consumption data.
- Import function
 - Functions for importing exported data into the archive tables in order to evaluate them using the applications listed above.

2 Data Management

Overview

Menu	System administration → Archiving → Data management
Transaction code	arccfg
Function authorization	arccfg.*

Purpose

Use this application to view or change the centrally managed [Archiving](#) settings for recorded data.

Integration

Use this application to configure the settings for archiving / data management for all components/functions.

Field descriptions

Product

Enter the HYDRA product group for which you want to define a rule in the *Product* field.

Object

Use the *Object* to define the data you want to retain.

Retention period

Use the fields for the *Retention period* to define how long data should be available until data is archived.

Unit

Specify the retention period in days, months or years.

Last retention date

This field is computed and specifies the last day when data is still available in this object.

Action

You can choose from three options to specify the processing for an object:

D = delete	Object is deleted
M = move (archive)	Object is transferred to the next area
X = export	Object is unloaded (XUNLOAD format) and then deleted

Target object

Currently, the target object automatically results from the detail configuration.

Condition

The characters in this field are added as an additional condition to the database command that controls the action. This condition is linked to the other conditions of the original command using AND and is set in parentheses.

Last run (date/time)

Indicates the point in time when this rule was applied last.

License

Indicates which license is required for archiving. You can enter several licenses (separated by a space). If none of these licenses is licensed in the system, the data is either deleted (action D) or unloaded into files (action M or X), depending on the relevant action.

Path

Optional path to generate the file export (unload). The archiver currently only supports local HYDRA server drives. If you do not enter a path, file exports are filed as follows:

<HYDRADIR>/<SYSTEM>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR ... HYDRA directory

SYSTEM ... System number

YYYY-MM-DD ... Archiving date (YYYY ... Year, MM ... Month, DD ... Day)

PRODUCT ... Product group from archiving configuration

Administration table

Name of the administration table where archiving logs are stored.

Administration duration (retention period of administration table)

Use the fields for the *retention period* to specify how long the logs should be available in the administration table before being deleted.

Unit

Specify the retention period in days, months or years.

Archiving step

Specifies if this archiving process uses archiving function I or II. Archiving function I: The function moves data from the online data set to archive tables or deletes the data: setting M (medium-term archive). Archiving function II: The function moves data from the archive table to the file export or deletes the data: setting L (long-term archive).

Configuration

Indicates whether or not the configuration is active. Possible values: Y/N

Archiving type

Identifier for time or object-related archiving. Supported modes: O = Object-related archiving (i.e. data is archived for each object individually). Z = Time-related archiving (i.e. data is archived without any object reference).

Note: Time and object-related archiving differ from each other significantly in the archiving performance (runtime). Object-related archiving of mass data is not recommended.

Priority

Integral value greater than 0. Indicates the processing sequence if several objects are defined for a product group. Processing starts with the lowest value.

Master table

Table including the data to be archived. The extensions entered in the *Condition* field refer to this table.

Date column

Date column in the master table. The system uses this date column to evaluate the retention period for the data to be archived.

Key 1

Unique key column in the master table; this key identifies the data to be archived. You can define up to 5 key columns for object-related archiving. Time-related archiving only supports one primary key in *Key 1*.

Keys 2 – 5

Additional optional key columns for object-related archiving.

Comment

Use the comment line to describe the archiving configuration.



If you copy the archiving configuration, the system currently only copies the data management configuration, but no defined data records from the object details. Currently, you must use the respective dialog to copy the data records of the object details to the new configuration.

Module	Version(s)	Description of archiving
BDE	8.1	See here
CAQ	8.1	See here
LLE	8.1	See here
MDE	8.1	See here
MPL	8.1	See here

PEP	8.1/8.2	See here
PZE/PZW	8.1/8.2	See here
ZKS	8.1/8.2	See here

3 CAQ-specific Configurations

3.1 Overview

3.1.1 Archiving of inspection requirements

3.1.1.1 Data structures

The default archiving of inspection requirements is object-related, i.e. each separate inspection requirement is evaluated in the archiving process. If the conditions specified via configuration parameters are fulfilled, the requirement is archived. In this process, all detailed data included is also archived.

Optionally, you can use the above parameters to delete the data (instead of archiving).

By default, there is a separate configuration for each individual HYDRA-CAQ data type (in-production inspection, goods receipt or goods issue inspection, initial sample inspection, QMS data). You can therefore archive inspection requirements of the goods receipt area in other intervals than data of the production data.

By default, the details saved for inspection requirements are identical for each data type. This may be the following data:

Data	Source table(s)	
Inspection requirements	caq_pruefanf caq_pan_zusatz	
Inspection orders	caq_paukop	
Inspection order configurations Characteristic configurations	caq_paukonf	
Inspection points	caq_numpool caq_ppktm_info	
QMS: Dynamic modification history of inspection points based on characteristics	caq_dyhis_ppktmm	
Characteristics	caq_merkmal caq_merk_zusatz	
Inspection frequencies	caq_prueffreq	
Inspection specifications depending on quantities	caq_mengabh_prf	
Documents	caq_dokus	
Tool assignments	caq_werkzzuord	
Samples Characteristic results	caq_paustich	
Assignment of samples to numbers Assignment of samples to inspection points	caq_paunumm	
Characteristic attributes	caq_paumm_ausp	
Single values	caq_paumwert	
Failure analysis entries	caq_fhlanal	
Measures and corresponding parameters	caq_massn	

Data	Source table(s)	
	caq_mass_param	
Inspection matrix Assignment of characteristics to inspection points	caq_pruefmatrix	

Events and logging entries for inspection requirements are archived separately. The archiving configuration of this data is described in the sections that follow.

Inspection requirements for the PMV data type (calibration/maintenance) are not archived by default.



The archiving configuration defines key fields. The key fields are used to filter the data when the data is reloaded. For example, all inspection requirements of the period x are filtered for article 4711. The key fields are assigned as follows by default.

Key 1: rec_type, e.g. FEP

Key 2: area, e.g. F for production

Key 3: pruefanf_nr (unique inspection requirement number)

Key 4: auftrag_nr (order number of inspection requirement)

Key 5: artikel_nr (article number of inspection requirement)

If you use the database Oracle, all 5 key fields must be filled for the archiving of the inspection requirements. If for example the article number is missing in an inspection requirement, this inspection requirement is not archived. If you want to archive such inspection requirements, the relevant key field must not have a content in the archiving configuration. With the example of the article number, this means that you cannot filter by the article number in a reload.

3.1.1.2 Standard configuration (with license FEP-/WEP-/QMS-ARC)

If one of the licenses FEP-/WEP-/QMS-ARC is available, the inspection requirements of the below mentioned data types are saved by default using the structure mentioned above.

For each of these data types, a separate configuration is provided. You can use the separate configurations to configure separate archiving periods, for example. The data type WAP (goods issue) is assigned to the licenses FEP-*

The standard configuration of the inspection requirements provides a two-step archiving.

In a first step, data is moved to the medium-term data area. Here, data is directly available for evaluations/reports using the medium-term data area. This data cannot be changed.

In a second step, data is moved to the long-term data area. In this case, data is no longer directly available for evaluations/reports. Data needs to be reloaded first to be able to use it for evaluations/reports in HYDRA.

Standard configurations for the first and second archiving level of inspection requirements and the respective intervals are described in the following table.

Product	Object	Description of the action	Default interval
CAQ	FEP	Moving the production inspection requirements from the online data set to the medium-term data set.	1 year
CAQ	A_FEP	Moving the production inspection requirements from the medium-term data set to the long-term data set.	3 years
CAQ	WEP	Moving the inspection requirements of the goods receipt from the online data set to the medium-term data set.	1 year
CAQ	A_WEP	Moving the inspection requirements of the goods receipt from the medium-term data set to the long-term data set.	3 years
CAQ	WAP	Moving the inspection requirements of the goods issue from the online data set to the medium-term data set.	1 year
CAQ	A_WAP	Moving the inspection requirements of the goods issue from the medium-term data set to the long-term data set.	3 years
CAQ	EMU	Moving the inspection requirements of initial samples from the online data set to the medium-term data set.	1 year
CAQ	A_EMU	Moving the inspection requirements of initial samples from the medium-term data set to the long-term data set.	3 years
QMS	QMS	Moving the QMS inspection requirements from the online data set to the medium-term data set.	3 months
QMS	A_QMS	Moving the QMS inspection requirements from the medium-term data set to the long-term data set.	3 years

To identify the intervals, the editing date of the inspection requirement is used as reference. Not all inspection requirements are archived. The respective status specifies whether an inspection requirement is archived or not. By default, only completed and canceled inspection requirements are archived.

QMS collective requirements must be uploaded to the PPS system in order to be archived. Administrative data on archived CAQ inspection requirements is archived for 12 years in the arc_verw_caq table by default.

3.1.1.3 Activities without license FEP-/WEP-/QMS-ARC

If no active license FEP-/WEP-/QMS-ARC is available, this archiving configuration does not affect the inspection requirements.

In contrast to other HYDRA data, inspection requirements are not removed in this case. They permanently remain in the online data area, unless they are included in another archiving configuration. The data type WAP (goods issue) is assigned to the licenses FEP-*.

The inspection requirements of QMS data are an exception. The results of these inspection requirements are normally uploaded to the PPS system. For this reason it is not required to keep this data in HYDRA. The data is removed from the system after expiration of the period of time defined for the product QMS/object QMS in section [1.1.1.2 Standard configuration \(with license FEP-/WEP-/QMS-ARC\)](#), just as it is the case for other HYDRA data areas if the configuration has not been changed.

3.1.2 Archiving of collective requirements

3.1.2.1 Data structures

The default archiving of collective requirements is object-related, i.e. each collective requirement is evaluated in the archiving process. If the conditions specified via configuration parameters are fulfilled, the requirement is archived. In this process, all detailed data included is also archived.

Optionally, you can use the above parameters to delete the data (instead of archiving).

In contrast to the archiving of inspection requirements, the collective requirements are not archived with reference to their data types. Consequently, it is not possible to define parameters for different intervals for the collective requirements of different data types.

The details saved by default for a collective requirement are:

Data	Source table(s)	
Collective requirements	caq_sammelanf	
Inspection frequencies	caq_prueffreq	

Logging entries for collective requirements are archived separately. The archiving configuration of this data is described in the sections that follow.

3.1.2.2 Standard configuration (with license FEP-/WEP-/QMS-ARC)

If one of the licenses FEP-/WEP-/QMS-ARC is available, the collective requirements are saved by default using the structure mentioned above and irrespective of their data type.

The standard configuration of the collective requirements provides a two-step archiving.

In a first step, data is moved to the medium-term data area. Here, data is directly available for evaluations/reports using the medium-term data area. This data cannot be changed.

In a second step, data is moved to the long-term data area. In this case, data is no longer directly available for evaluations/reports. Data needs to be reloaded first to be able to use it for evaluations/reports in HYDRA.

The standard configurations for the first and second archiving level of the collective requirements are described below.

Product	Object	Description of the action	Default interval
CAQ	SAN	Moving collective requirements from the online data set to the medium-term data set.	1 year
CAQ	A_SAN	Moving collective requirements from the medium-term data set to the long-term data set.	3 years

To identify the intervals, the editing date of the collective requirement is used as reference.

By default, only the collective requirements without inspection requirements in the online data area are archived in the medium-term data area. This ensures that a collective requirement is only archived in the medium-term data area, if all included inspection requirements have also been archived.

Archiving to the long-term data area works on the same principle. Only if all inspection requirements included in the collective requirement do no longer exist in the medium-term data set, the collective requirement is transferred to the long-term data set (respecting the interval).

Administrative data on archived CAQ collective requirements is archived for 12 years in the arc_verw_caq table by default.

3.1.2.3 Activities without license FEP-/WEP-/QMS-ARC

If no active license FEP-/WEP-/QMS-ARC is available, this archiving configuration does not affect the collective requirements.

In contrast to other HYDRA data, collective requirements are not removed in this case. They permanently remain in the online data area, unless they are included in another archiving configuration.

3.1.3 Archiving of CAQ events

3.1.3.1 Data structures

By default, CAQ events are archived in relation to time. The archiving is always performed and it does not matter if the included CAQ objects (e.g. inspection requirements or their details) have been archived or not.

Optionally, you can use the above parameters to delete the data (instead of archiving).

By default, the events are archived separately, if applicable with the relevant detail data. The following data is archived:

Data	Source table(s)
CAQ events	event_caq
Optional dialog data for events	event_dlg_data

3.1.3.2 Standard configuration (with license FEP-/WEP-/QMS-ARC)

If one of the licenses FEP-/WEP-/QMS-ARC is available, the CAQ events are saved by default using the structure mentioned above and irrespective of their data type.

The standard configuration of the CAQ events provides a two-step archiving.

In a first step, data is moved to the medium-term data area. Here, data is directly available for evaluations/reports using the medium-term data area.

In a second step, data is moved to the long-term data area. In this case, data is no longer directly available for evaluations/reports. Data needs to be reloaded first to be able to use it for evaluations/reports in HYDRA.

The standard configurations for the first and second archiving level of the CAQ events are described below.

Product	Object	Description of the action	Default interval
CAQ	EREIGCAQ	Moving entries of CAQ events from the online data set to the medium-term data set.	35 days
CAQ	A_ EREIGCAQ	Moving entries of CAQ events from the medium-term data set to the long-term data set.	3 years

To identify the intervals, the date of the CAQ event is used as reference.

Administrative data on archived CAQ events is archived for 12 years in the `arc_verw_caq` table by default.

3.1.3.3 Activities without license FEP-/WEP-/QMS-ARC

If no active license FEP-/WEP-/QMS-ARC is available, this archiving configuration does not affect the CAQ events.

In contrast to other HYDRA data, CAQ events are not removed in this case. They permanently remain in the online data area, unless they are included in another archiving configuration.

3.1.4 Archiving of CAQ logging entries

3.1.4.1 Data structures

By default, CAQ logging entries are archived in relation to time. The archiving is always performed and it does not matter if the included CAQ objects (e.g. inspection requirements or their details) have been archived or not.

Optionally, you can use the above parameters to delete the data (instead of archiving).

By default, the logging entries are archived separately, if applicable with the relevant detail data. The following data is archived:

Data	Source table(s)
Logging entries	hyd_logging
Additional data to logging entries	hyd_logging_data

3.1.4.2 Standard configuration (with license FEP-/WEP-/QMS-ARC)

If one of the licenses FEP-/WEP-/QMS-ARC is available, the CAQ logging entries are saved by default using the structure mentioned above.

The standard configuration of the CAQ logging entries provides a two-step archiving.

In a first step, data is moved to the medium-term data area. Here, data is directly available for evaluations/reports using the medium-term data area.

In a second step, data is moved to the long-term data area. In this case, data is no longer directly available for evaluations/reports. Data needs to be reloaded first to be able to use it for evaluations/reports in HYDRA.

The default configurations for the first and second archiving level of CAQ logging entries are described in the below table.

Product	Object	Description of the action	Default interval
CAQ	LOG	Moving CAQ logging entries from the online data set to the medium-term data set.	35 days
CAQ	A_LOG	Moving the CAQ logging entries from the medium-term data set to the long-term data set.	3 years

To identify the intervals, the date of the logging entry is used as reference.

Administrative data on archived CAQ logging entries is archived for 12 years in the arc_verw_caq table by default.

3.1.4.3 Activities without license FEP-/WEP-/QMS-ARC

If no active license FEP-/WEP-/QMS-ARC is available, this archiving configuration does not affect the CAQ logging entries.

In contrast to other HYDRA data, CAQ logging entries are not removed in this case. They permanently remain in the online data area, unless they are included in another archiving configuration.

3.1.5 Archiving of the document management

The documents of the HYDRA document management are archived when the relevant object is archived. The files themselves remain at their original storage location and are not archived.

As part of the HYDRA document management of the CAQ, you can assign documents to the following objects.

- Inspection points
- Inspection step characteristics/inspection point characteristics
- Measured values/attributive inspection results

To archive the documents, make the following entries in the data management.

Product	Object	Description of the action	Default interval
CAQ	DOCLINK	Moving the document entry from the online data set to the medium-term data set.	0 days

Product	Object	Description of the action	Default interval
CAQ	A_DOCLINK	Moving the document entry from the medium-term data set to the long-term data set.	0 days

Set the interval to "0 days" so that the archiving is directly performed after the archiving of the respective object.

4 Reload Manager

Summary

HYDRA menu	System administration → Archiving → Reload manager
FEDRA menu	System administration → Archiving → Reload manager
Transaction code	arcrlid
Function authorization	arcrlid arcrlid.export arcrlid.import

Purpose

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evaluate restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.



The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- Cyclic

Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.

Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is calculated subject to the configuration and entered in the reload management table.

- User-specific

Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.

When data is imported, the `HYD_REL_MANAGEMENT.DELETE_DATE` field is computed subject to the configuration and entered in the reload management table.

- Manual

Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.